ADS-B Flight Reconstruction in FAA-Managed Airspace

Jonah G github.com/PlaneQuery/planequery-flights-algorithm

Abstract

We present an algorithm for deriving flights from ADS-B data using an LLM-generated rule-based state machine for flight segmentation and a LightGBM model for takeoff and landing airport identification. The method is evaluated against ADS-B-validated subsets of FAA SWIM Flight Data Publication Service (SFDPS) and Bureau of Transportation (BTS) flight data. It is compared against existing algorithms from OpenSky and ADS-B Exchange.

Using ADS-B Exchange ADS-B data, the proposed algorithm achieved an F1 score of 0.997 against SFDPS and 1.000 against BTS, with corresponding airport identification accuracies of 99.45% and 100% among correctly segmented flights. Against the SFDPS evaluation set, the same segmentation method achieved F1 scores of 0.986 with ADSB.lol data and 0.963 with OpenSky data, demonstrating the effect of differences in ADS-B network quality. The proposed method outperformed ADS-B Exchange and OpenSky algorithms, which achieved SFDPS flight-segmentation F1 scores of 0.960 and 0.930, respectively.

The LightGBM airport model provided its largest gains on the weaker ADS-B sources: relative to a nearest-airport method, airport accuracy increased from 91.71% to 94.97% with ADSB.lol data and from 81.19% to 88.62% with OpenSky data. With ADS-B Exchange data, airport accuracy increased marginally from 99.02% to 99.45%.

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I. Evaluation Datasets

FAA SWIM SFPDS

The FAA SWIM SFDPS feed was recorded in real-time for 4 months using Faa-Swim-Jumpstart. An algorithm (src/data_engineering/flights/sfdps_to_flights.py) identifies flights from the SFPDS data feed.

Issues with data: Flights outside of US FAA airspace are not present. Flights that fly VFR (Visual Flight Rules) and do not file flight plans are not present. Some government/military broadcast on ADS-B but are not present in SFDPS. There are errors in data including incorrectly entered airports and incorrect flights.

Bureau of Transportation Statistics (BTS)

Issues: Flights outside of US FAA airspace are not present. Same plane can have a flight with a landing time that is after the next flights takeoff time (overlapping flights).

Making Gold datasets from the test sets

Due to errors and missing flights from the datasets sets we ran an algorithm on datasets that verifies that flights match ADS-B data. Many correct flights that have poor ADS-B coverage are eliminated. If flights do not match ADS-B for any portion of the day, entire icao is eliminated (src/flights/flights_match_adsb.py).

Table 1. Flight Coverage (2026-03-01)

Flights Source	ADS-B Source	Total Flights	Gold Flights	Flights Kept
SFDPS	ADSBExchange	27,082	6,735	24.87%
SFDPS	ADSB.lol	27,082	6,375	23.54%
SFDPS	OpenSky	27,082	3,206	11.84%
BTS	ADSBExchange	17,609	4,218	23.95%
BTS	ADSB.lol	17,609	3,905	22.18%
BTS	OpenSky	17,609	667	3.79%

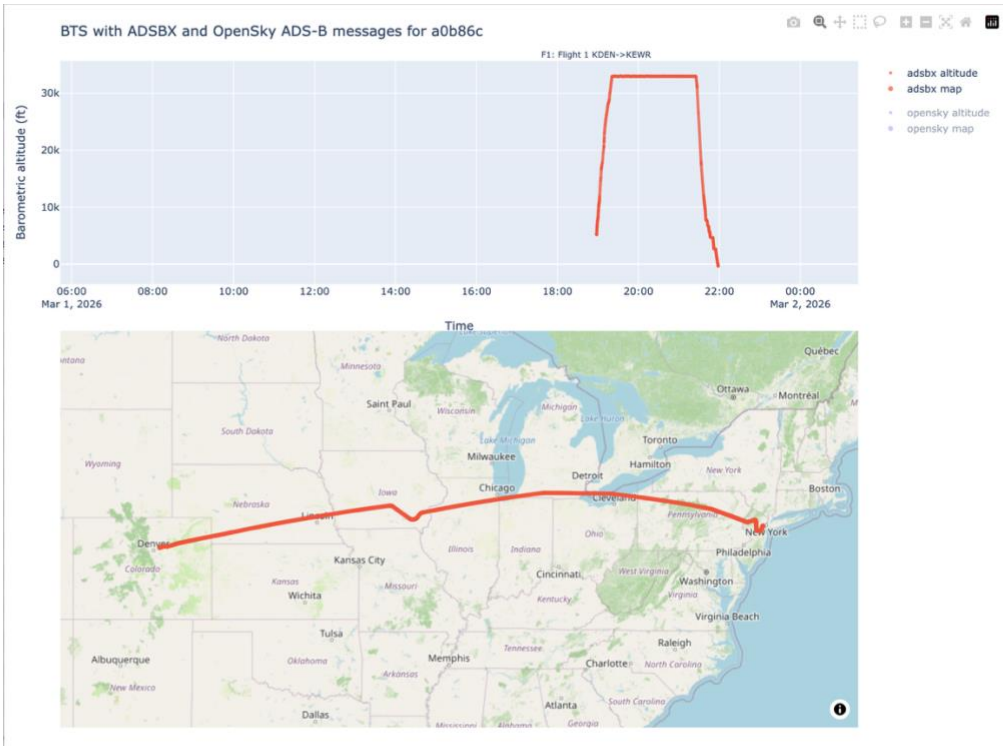
Table 2. Aircraft Coverage (2026-03-01)

Flights Source	ADS-B Source	Total ICAOs	Gold ICAOs	ICAOs Kept
SFDPS	ADSBExchange	12,396	3,162	25.51%
SFDPS	ADSB.lol	12,396	2,947	23.77%
SFDPS	OpenSky	12,396	1,741	14.04%
BTS	ADSBExchange	4,718	1,017	21.56%
BTS	ADSB.lol	4,718	953	20.20%
BTS	OpenSky	4,718	191	4.05%

Generated by scripts/testset_to_gold_elimination.py

The percentage kept is indicative of the ADS-B network coverage quality. ADS-B Exchange is better than ADSB.lol. OpenSky has errant lat/lon points that cause algorithm to produce false negatives. Since ADS-B Exchange network has the best data, it was used for gold datasets in the following sections.

OpenSky example: a0b86c errant lat/lon points in OpenSky causes algorithm to output false negative, while algorithm run on ADS-B Exchange data correctly keeps it.



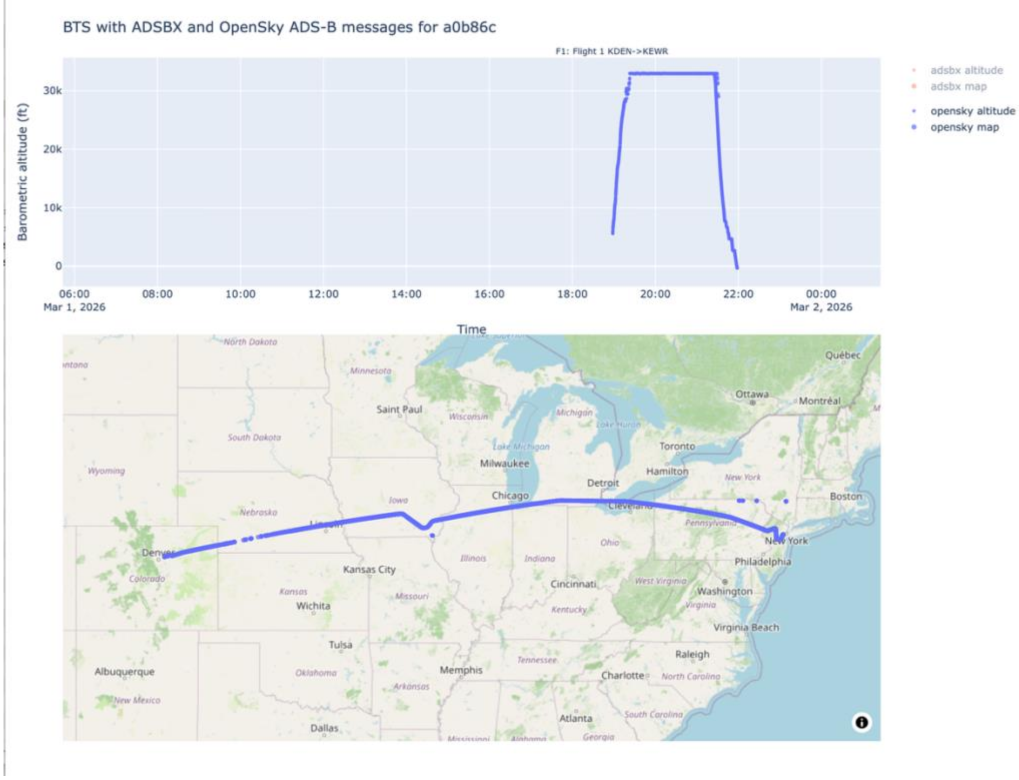


Figure [*https://globe.adsbexchange.com/?icao=a0b86c&lat=27.561&lon=-97.234&zoom=4.3&showTrace=2026-03-01&leg=1&timestamp=1772392500*](https://globe.adsbexchange.com/?icao=a0b86c&lat=27.561&lon=-97.234&zoom=4.3&showTrace=2026-03-01&leg=1&timestamp=1772392500)

II. Algorithm

Flight Segmentation

We initially developed a hand-coded state-machine algorithm based on ground speed. We then created a pytest-based evaluation harness using SFDPS data, restricted to LADD and PIA aircraft. Then prompted Codex in VSCode with:

using the tests

((.venv)) jonahgoode@jonahs-MacBook-Air planequery-data % pytest

tests/flights/evaluation/test_flights_algorithm.py -s

[...]

and notebooks/flights/flights_algorithm_datascience.ipynb if applicable. Improve src/flights_algorithm/main.py for pia and ladd flights you should be able to get it above 0.99 F1. You can ignore airport matching. You should be able to greatly simplify the algorithm. Be careful of doing anything that would extremely overfit it to the test set.

ChatGPT 5.5 Extra High thinking ran for 1h 7 min (paper/flight-segmenation-llm-output)

Airports

A LightGBM model using airports from ourairports.com and the below features:

```
FEATURE_NAMES = [
    "distance_km",
    "elevation_above_airport_ft",
    "aircraft_category",
    "airport_type",
    "bearing_airport_to_point_deg",
    "bearing_point_to_airport_deg",
    "track_error_deg",
    "direction_error_deg",
    "along_track_km",
    "cross_track_km",
    "candidate_in_expected_direction",
]
```

All features are relative to a given airport. No absolute data such as lat, lon of the airplane or airport is used to avoid overfitting and allow generalization. Gold SFDPS data and ADSB.lol ADS-B data was used to train model with data starting from 2026-06-01.

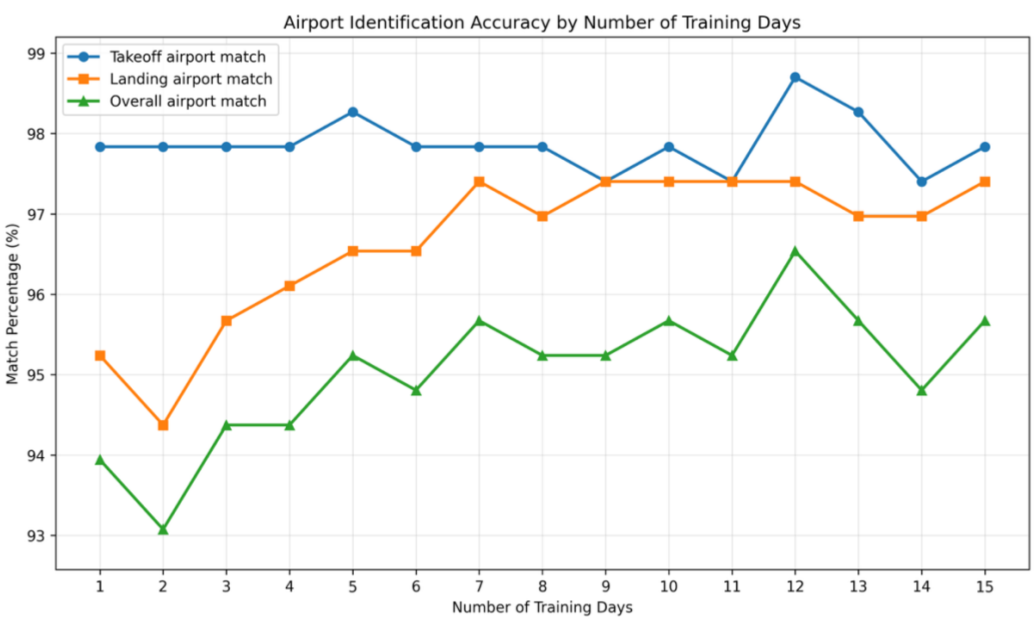


Figure 1 Evaluated on 2026-02-01 SFDPS Gold Data. Generated by scripts/lightgbm_model_performance_vs_days.py

Model with 12 days, 2026-06-01 -> 2026-06-12 had highest accuracy therefore this model was used.

III. Results

For comparison, results for “Algorithm, no airport model” are also provided. This approach uses the same flight segmentation algorithm, but instead of the LightGBM model, the takeoff and landing airports are assigned as the airports closest to the latitude and longitude of the first and final ADS-B messages in each flight segment, respectively.

Gold Datasets

Table 3 Flight Segmentation Metrics

ADS-B Data	Method	Gold Data	TP	FP	FN	Precision	Recall	F1
ADSB.lol	Algorithm	SFDPS	6,602	65	124	99.03%	98.16%	98.59%
ADSB.lol	Algorithm	BTS	3,513	19	90	99.46%	97.50%	98.47%
ADSBExchange	Algorithm	SFDPS	6,716	23	16	99.66%	99.76%	99.71%
ADSBExchange	Algorithm	BTS	3,616	0	0	100.00%	100.00%	100.00%
OpenSky	Algorithm	SFDPS	6,371	136	352	97.91%	94.76%	96.31%
OpenSky	Algorithm	BTS	3,500	39	112	98.90%	96.90%	97.89%
ADSB.lol	Algorithm, no airports	SFDPS	6,611	65	117	99.03%	98.26%	98.64%
ADSB.lol	Algorithm, no airports	BTS	3,521	22	83	99.38%	97.70%	98.53%
ADSBExchange	Algorithm, no airports	SFDPS	6,716	23	16	99.66%	99.76%	99.71%
ADSBExchange	Algorithm, no airports	BTS	3,616	0	0	100.00%	100.00%	100.00%
OpenSky	Algorithm, no airports	SFDPS	6,373	136	350	97.91%	94.79%	96.33%
OpenSky	Algorithm, no airports	BTS	3,500	39	112	98.90%	96.90%	97.89%
—	OpenSky	SFDPS	6,041	219	689	96.50%	89.76%	93.01%
—	OpenSky	BTS	3,526	68	90	98.11%	97.51%	97.81%
—	ADSBExchange	SFDPS	6,244	31	484	99.51%	92.81%	96.04%
—	ADSBExchange	BTS	3,484	1	129	99.97%	96.43%	98.17%

Table 4 Airport Identification Metrics

Computed only for flights where segmentation matches gold data.

ADS-B Data	Method	Gold Data	Takeoff Match	Landing Match	Both Airports Match
ADSB.lol	Algorithm	SFDPS	96.83%	97.80%	94.97%
ADSB.lol	Algorithm	BTS	98.98%	99.32%	98.29%
ADS-B Exchange	Algorithm	SFDPS	99.81%	99.55%	99.45%
ADS-B Exchange	Algorithm	BTS	100.00%	100.00%	100.00%
OpenSky	Algorithm	SFDPS	93.78%	94.26%	88.62%
OpenSky	Algorithm	BTS	97.40%	97.03%	94.43%
ADSB.lol	Algorithm, no airports	SFDPS	95.61%	95.67%	91.71%
ADSB.lol	Algorithm, no airports	BTS	97.98%	97.87%	95.85%
ADS-B Exchange	Algorithm, no airports	SFDPS	99.67%	99.30%	99.02%
ADS-B Exchange	Algorithm, no airports	BTS	100.00%	99.83%	99.83%
OpenSky	Algorithm, no airports	SFDPS	91.10%	88.69%	81.19%
OpenSky	Algorithm, no airports	BTS	95.49%	94.51%	90.00%
—	OpenSky	SFDPS	71.49%	66.71%	49.33%
—	OpenSky	BTS	82.87%	74.48%	61.68%

ADS-B Data	Method	Gold Data	Takeoff Match	Landing Match	Both Airports Match
—	ADSBExchange	SFDPS	99.86%	99.90%	99.76%
—	ADSBExchange	BTS	100.00%	100.00%	100.00%

Results Using Unfiltered SFDPS and BTS Data

Table 5 Flight Segmentation Metrics

ADS-B Data	Method	Gold Data	TP	FP	FN	Precision	Recall	F1
ADSB.lol	Algorithm	SFDPS	25,845	7,141	1,187	78.35%	95.61%	86.12%
ADSB.lol	Algorithm	BTS	14,801	1,229	542	92.33%	96.47%	94.36%
ADSBExchange	Algorithm	SFDPS	26,453	8,639	607	75.38%	97.76%	85.12%
ADSBExchange	Algorithm	BTS	15,297	1,872	83	89.10%	99.46%	93.99%
OpenSky	Algorithm	SFDPS	24,661	7,407	2,359	76.90%	91.27%	83.47%
OpenSky	Algorithm	BTS	14,772	1,720	593	89.57%	96.14%	92.74%
ADSB.lol	Algorithm, no airports	SFDPS	25,908	7,892	1,133	76.65%	95.81%	85.17%
ADSB.lol	Algorithm, no airports	BTS	14,831	1,623	514	90.14%	96.65%	93.28%
ADSBExchange	Algorithm, no airports	SFDPS	26,453	8,639	607	75.38%	97.76%	85.12%
ADSBExchange	Algorithm, no airports	BTS	15,297	1,872	83	89.10%	99.46%	93.99%
OpenSky	Algorithm, no airports	SFDPS	24,669	7,415	2,351	76.89%	91.30%	83.48%
OpenSky	Algorithm, no airports	BTS	14,774	1,724	591	89.55%	96.15%	92.73%
—	OpenSky	SFDPS	23,796	6,699	3,248	78.03%	87.99%	82.71%
—	OpenSky	BTS	14,849	1,784	531	89.27%	96.55%	92.77%
—	ADSBExchange	SFDPS	24,245	5,531	2,761	81.42%	89.78%	85.40%
—	ADSBExchange	BTS	14,830	1,150	540	92.80%	96.49%	94.61%

Table 6 Airport Identification Metrics

ADS-B Data	Method	Gold Data	Takeoff Match	Landing Match	Both Airports Match
ADSB.lol	Algorithm	SFDPS	92.78%	90.87%	86.83%
ADSB.lol	Algorithm	BTS	98.94%	99.30%	98.24%
ADSBExchange	Algorithm	SFDPS	96.00%	92.04%	90.60%
ADSBExchange	Algorithm	BTS	99.92%	99.93%	99.86%
OpenSky	Algorithm	SFDPS	90.95%	89.26%	83.48%
OpenSky	Algorithm	BTS	97.69%	97.34%	95.03%
ADSB.lol	Algorithm, no airports	SFDPS	91.26%	89.19%	83.81%
ADSB.lol	Algorithm, no airports	BTS	98.15%	98.05%	96.20%
ADSBExchange	Algorithm, no airports	SFDPS	95.76%	91.79%	90.17%
ADSBExchange	Algorithm, no airports	BTS	99.92%	99.80%	99.72%
OpenSky	Algorithm, no airports	SFDPS	88.17%	85.21%	77.52%
OpenSky	Algorithm, no airports	BTS	96.32%	95.17%	91.50%
—	OpenSky	SFDPS	72.29%	66.78%	50.63%
—	OpenSky	BTS	83.40%	75.35%	62.71%
—	ADSBExchange	SFDPS	97.02%	94.07%	93.00%
—	ADSBExchange	BTS	99.97%	99.97%	99.93%

Table 7 ChatGPT 5.6 Sol (medium) analysis of flight segmentation mismatches

Dataset	Test set Error	Gold-Dataset Error	Error in both
Algorithm ADS-B Exchange	17.39%	78.26%	4.35%
Algorithm ADSB.lol	78.95%	18.95%	2.11%
ADS-B Exchange	97.61%	2.15%	0.24%
OpenSky	96.77%	0.51%	2.72%

Algorithm ADS-B Exchange Mismatch Analysis

Summary

Manual review classified the 23 mismatched ICAOs as follows:

- Gold-dataset/df1 errors: 18 ICAOs
- Genuine algorithm/df0 failures: 4 ICAOs
- Errors in both datasets: 1 ICAO (`a654ec`)

Manual Review Findings

ICAO	Incorrect Dataset	Manual Review Finding
`a25dca`	df1 gold	SFDPS combines the KSUS–KUBX–KSUS activity into a single KSUS–KSUS flight.
`a3a1dc`	df1 gold	The second flight departed KRBD at approximately 21:10, whereas df1 incorrectly reuses the 18:11 departure time.
`a1e6b9`	df1 gold	The aircraft remained on the ground at KHOU until the actual takeoff at approximately 13:37, whereas df1 reports a 12:34 departure.
`adab24`	df0 algorithm	The algorithm incorrectly identifies an FA71 stop during gaps in ADS-B coverage. The aircraft remained airborne and did not come closer than approximately 42 km to FA71.
`a81525`	df0 algorithm	The algorithm completely misses a clear KAUS ground–air–ground flight that occurred from approximately 16:56 to 17:21.

ICAO	Incorrect Dataset	Manual Review Finding
a875ab	df1 gold	The KBKL–KLNS flight departed at approximately 10:38, whereas df1 incorrectly reuses the 06:12 departure time.
adccd0	df1 gold	The KPHL–KHPN flight departed at approximately 17:43, whereas df1 incorrectly reuses the 14:03 departure time.
a63c0b	df1 gold	The KENW–KBNA flight departed at approximately 16:25, whereas df1 incorrectly reuses the 12:48 departure time.
a1c2e3	df1 gold	The aircraft remained on the ground at KAPF until approximately 20:33, whereas df1 reports an 18:19 departure.
a1a4a7	df1 gold	A genuine ground stop at KNEW separates the KBHM–KNEW and KNEW–KBHM flights, whereas df1 combines them.
a5a744	df1 gold	The trajectory supports an intermediate landing or touch-and-go at KM89, whereas df1 combines the two flight legs.
a32756	df1 gold	SFDPS omits genuine morning pattern activity at KRIC. Although the algorithm's segmentation is imperfect, the flight identified as an additional flight is genuine.
a3e348	df1 gold	The FA54–KHOU flight departed at approximately 17:57, whereas df1 incorrectly reuses the 15:07 departure time.
a1d684	df1 gold	An intermediate stop at KTYR separates two flight legs, whereas df1 combines them into a single KACT–KACT flight.
a1492e	df1 gold	SFDPS omits a genuine local KCRG flight that occurred from approximately 18:51 to 20:01.
a654ec	Both	df1 omits the earlier local KFFZ flight, while df0 also fails to separate the later KFFZ–KPRC–KFFZ touch-and-go activity.
abfc78	df1 gold	The trajectory supports an intermediate stop or touch-and-go at KAQX, whereas df1 combines the KSAV–KAQX–KSAV activity.
aafb20	df1 gold	SFDPS omits a genuine local KDTO flight that occurred from approximately 17:11 to 17:48.

ICAO	Incorrect Dataset	Manual Review Finding
a25ec5	df0 algorithm	The algorithm misses the KACT touch-and-go and combines the KFTW–KACT–KFTW activity into one flight.
a6ea99	df1 gold	The raw data show that the aircraft remained at KGYY until the takeoff at approximately 16:36, whereas df1 reports a 15:30 departure.
ad963b	df1 gold	ADS-B data show the aircraft landing at KAPF at approximately 21:42, whereas df1 reports a 22:43 landing.
acb4b2	df0 algorithm	The algorithm misses the KUXL stop during a gap in ADS-B coverage and combines the KMSY–KUXL–KBTR activity into one flight.
a9b2c8	df1 gold	SFDPS omits a genuine local KFMY flight that occurred from approximately 17:28 to 18:37.

The classifications for `a25dca`, `a5a744`, `abfc78`, and `acb4b2` have lower confidence because the intermediate airport event occurred partly within a gap in ADS-B coverage. However, the surrounding approach and departure geometry supports the classifications shown above.

ADS-B Exchange Mismatch Analysis

Summary

The comparison produced 6,244 true positives, 31 false positives, and 484 false negatives. In total, 418 ICAO addresses were associated with mismatches, representing 515 mismatched rows.

Manual review classified the mismatches as follows:

- Confirmed df1/gold-dataset errors: 9 ICAOs
- Errors in both sources: 1 ICAO
- adsbx failures: 408 ICAOs

Most adsbx failures involved missing flights. A smaller number involved combining multiple SFDPS flight legs into a single long flight or truncating international or low-coverage flights. The `mismatch_classification` DataFrame contains the classification assigned to every mismatched ICAO.

Confirmed df1/Gold-Dataset Errors

ICAO	Review Finding
`a1a4a7`	adsbx correctly separates the KBHM–KNEW and KNEW–KBHM flights, whereas df1 combines them.
`a1c2e3`	The actual KAPF departure occurred at approximately 20:33, rather than 18:19 as recorded in df1.
`a1d684`	adsbx correctly separates the KACT–KTYR and KTYR–KACT flights, whereas df1 combines them.
`a3a1dc`	The KRBD–KMKY flight departed at approximately 21:10, whereas df1 incorrectly reuses the 18:11 departure time.
`a3e348`	The FA54–KHOU flight departed at approximately 17:57, whereas df1 incorrectly reuses the 15:07 departure time.
`a63c0b`	The KENW–KBNA flight departed at approximately 16:25, whereas df1 incorrectly reuses the 12:48 departure time.
`a6ea99`	The actual KGYY departure occurred at approximately 16:36, rather than 15:30 as recorded in df1.
`a875ab`	The KBKL–KLNS flight departed at approximately 10:38, whereas df1 incorrectly reuses the 06:12 departure time.
`adccd0`	The KPHL–KHPN flight departed at approximately 17:43, whereas df1 incorrectly reuses the 14:03 departure time.

Errors in Both Sources

For ICAO `a1e6b9`, df1 contains an incorrect departure time for the KHOU–KDAL flight, while adsbx also omits the subsequent KDAL–KSTL flight.

ADS-B Exchange and OpenSky flights data is significantly worse in quality than the SFDPS Gold dataset. However, the algorithm with ADS-B Exchange ADS-B data surpasses the quality of the SFDPS Gold dataset. Many of these mismatches are not because the algorithm is wrong but because the gold data set is wrong.

Comparison with FAA Daily Flight Estimates

According to faa.gov/air_traffic/by_the_numbers/air-traffic-by-the-numbers-FY2024.pdf there are, on average, 40,000 VFR flights and 44,000 IFR flights, for a total of approximately 84,000 flights per day in FAA-managed airspace.

Table 8 Flights with takeoff or landing airport in FAA-managed airspace

ADS-B source	Flights Source	FAA-managed airspace Flights	Difference	Percentage of FAA estimate
ADSB.lol	algorithm	67,464	-16,536	80%
OpenSky	algorithm	56,436	-27,564	67%
ADS-B Exchange	algorithm	70,462	-13,538	84%
OpenSky	OpenSky	53,282	-30,718	63%
ADS-B Exchange	ADS-B Exchange	50,434	-33,566	60%

Generated by notebooks/algoirthms_vs_faa_flight_total.ipynb

ADSBX algorithm failures examples

A004E9 Algorithm misses flight with good continental USA coverage

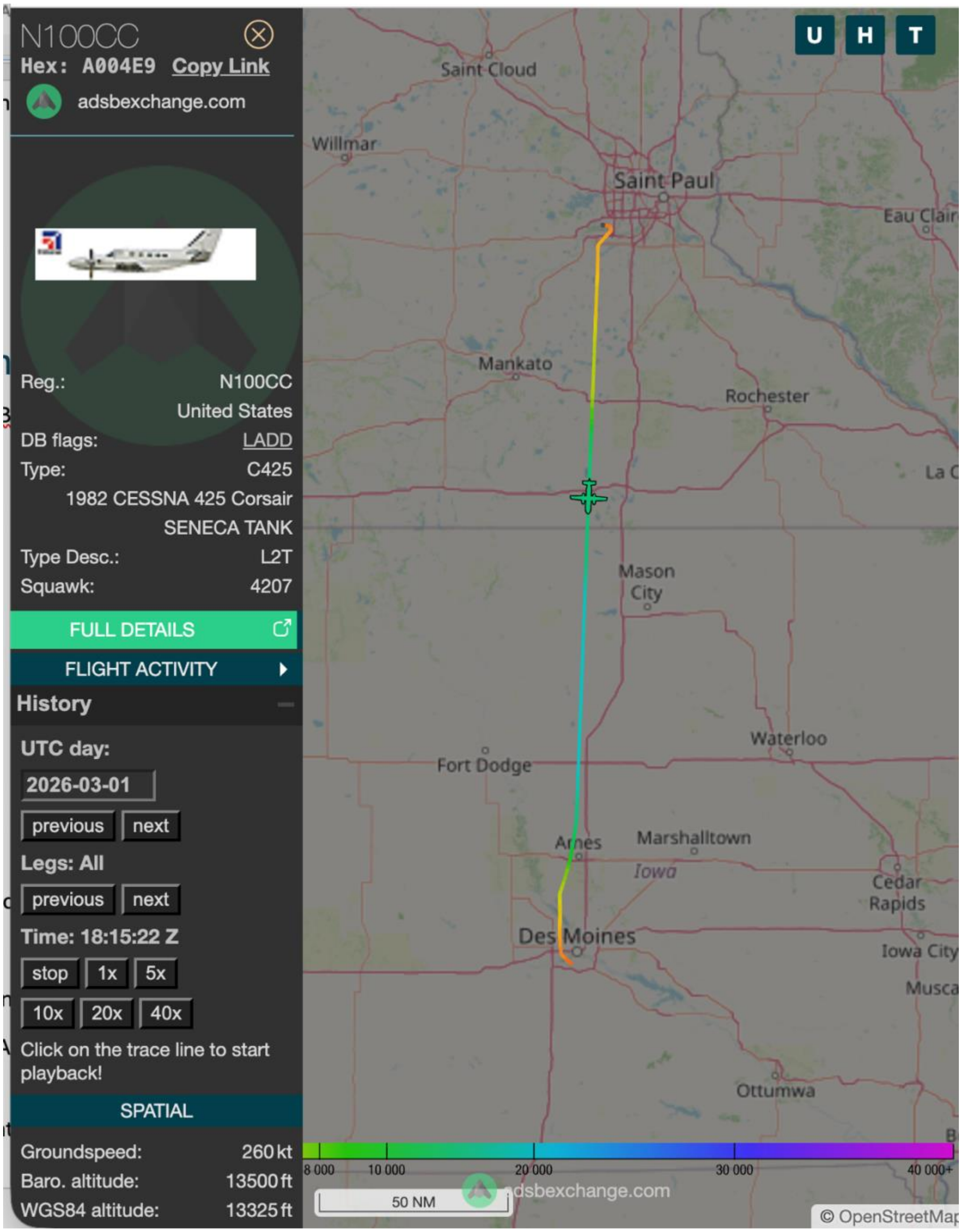
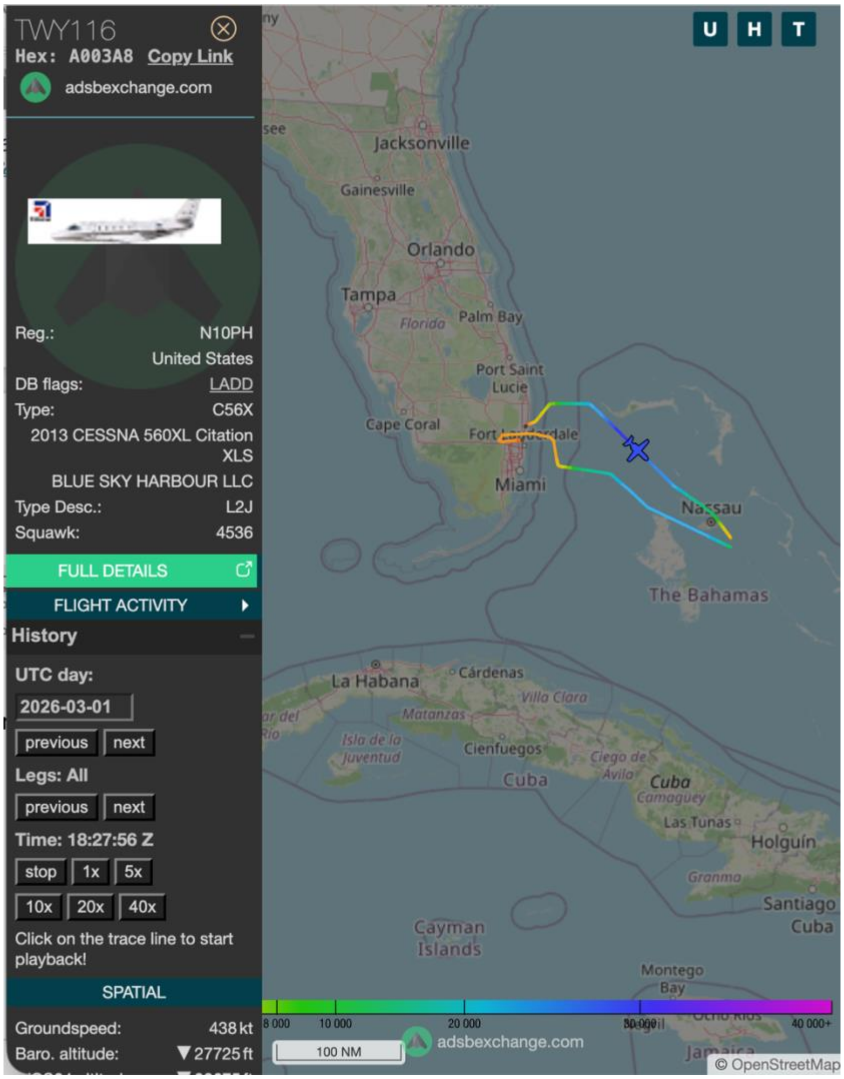


Figure flights produced by ADSB-Exchange algorithm for A00E49 on 2026-03-01.
https://globe.adsbexchange.com/?icao=a004e9&lat=43.098&lon=-93.829&zoom=7.3&showTrace=2026-03-01&timestamp=1772388923

A003A8: Algorithm does 1 flight instead of 2 for Island flight.



https://globe.adsbexchange.com/?icao=a003a8&lat=24.842&lon=-80.897&zoom=6.2&showTrace=2026-03-01&timestamp=1772389677

shape: (3, 10)

icao	takeoff_time	landing_time	takeoff_airport_ident	landing_airport_ident	takeoff_time_df1	landing_time_df1	takeoff_airport_ident_df1	landing_airport_ident_df1	match_status
str	datetime[ms]	datetime[ms]	str	str	datetime[ms]	datetime[ms]	str	str	str
"a003a8"	null	null	null	null	2026-03-01 18:08:00	2026-03-01 18:59:00	"KBCT"	"MYEN"	"df1_only"
"a003a8"	null	null	null	null	2026-03-01 19:54:00	2026-03-01 20:52:00	"MYEN"	"KFXE"	"df1_only"
"a003a8"	2026-03-01 18:08:13	2026-03-01 20:52:44	"KBCT"	"KFXE"	null	null	null	null	"df0_only"

Left is ADSB-Exchange algorithm flights. Right, _df1 columns are SFDPS gold flights.

OpenSky algorithm failure examples

a0103b extra flight

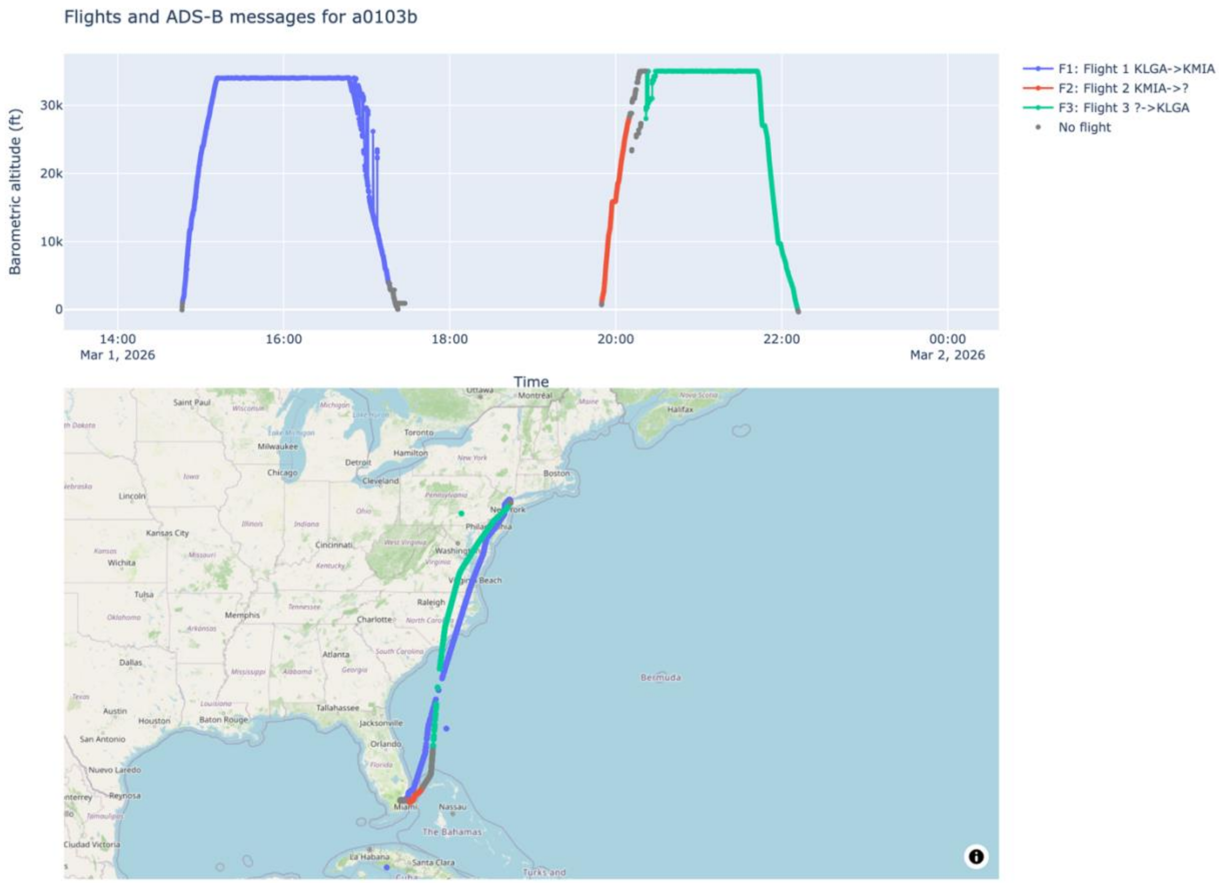


Figure 2 OpenSky ADS-B and flights data

shape: (3, 5)

icao24	firstSeen	lastSeen	estDepartureAirport	estArrivalAirport
str	i32	i32	str	str
"a0103b"	1772376406	1772385351	"KLGA"	"KMIA"
"a0103b"	1772394591	1772395798	"KMIA"	null
"a0103b"	1772396506	1772403096	null	"KLGA"

Figure Data from https://s3.opensky-network.org/data-samples/trino-tables/flights/day%3D1772323200/part-00001-a1c89589-f077-4615-bd84-d86ac2fc1497.c000.snappy.parquet

IV. Conclusions and Future Work

For high-quality ADS-B networks such as ADS-B Exchange, larger and more challenging labeled evaluation datasets are needed to distinguish remaining algorithm errors from errors in the reference data and to measure performance more precisely. For networks with worse coverage, the results indicate that further improvements to the segmentation algorithm could produce meaningful gains.

The airport-identification model also has substantial room for improvement. Potential directions include using neural networks and incorporating absolute geographic features in addition to the current airport-relative features.